

## Net change



- 1 In a closed habitat, the population of kangaroos  $P(t)$  is known to increase according to the function:

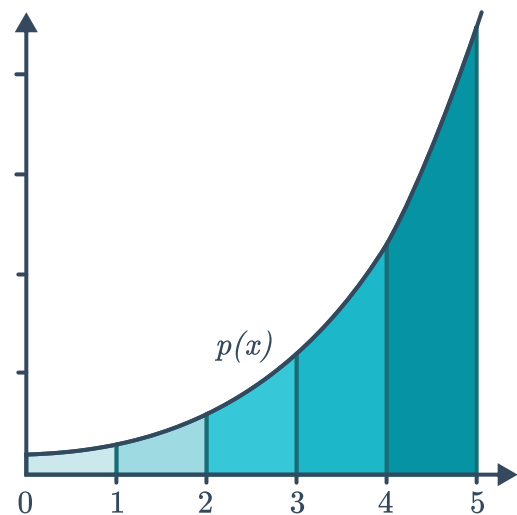
$$P'(t) = \frac{t}{2} + 9$$

Where  $t$  is measured in months since counting began.

- Calculate the total change in the population of kangaroos in the first 4 months since counting began.
- Find the number of months it will take from when counting began for the population of kangaroos to increase by 88.

- 2 The graph shows the population growth rate of fish which is given by  $p(x)$ . The total population growth of fish over  $n$  years is given by  $\int_0^n p(x) dx$  as shown in the table below:

| Year | Total growth |
|------|--------------|
| 1    | 1000         |
| 2    | 4000         |
| 3    | 13 000       |
| 4    | 40 000       |
| 5    | 121 000      |



- Determine whether the following represent the total population growth in year 2.
  - $\int_2^5 p(x) dx$
  - $\int_1^2 p(x) dx$
  - $\int_0^3 p(x) dx - \int_0^2 p(x) dx$
  - $\int_0^2 p(x) dx - \int_0^1 p(x) dx$
- Calculate the total population growth in year 2.
- Determine whether the following represent the total population growth from year 2 to 4.
  - $\int_2^4 p(x) dx$
  - $\int_0^4 p(x) dx - \int_0^2 p(x) dx$
  - $\int_0^2 p(x) dx - \int_0^4 p(x) dx$
  - $\int_4^2 p(x) dx$
- Calculate the total population growth from year 2 to 4.

- 3 An object is cooling and its rate of change of temperature, after  $t$  minutes, is given by:

$$T' = -10e^{-\frac{t}{5}}$$

- a Find the instantaneous rate of change of the temperature after 8 minutes, to two decimal places.
- b Find the total change of temperature after 9 minutes, to two decimal places.
- c Hence, find the average change in temperature over the first 9 minutes, to two decimal places.

- 4 The average rate of change of function  $f(x)$  over the domain  $a \leq x \leq b$  is given by:

$$\frac{\int_a^b f'(x) dx}{b - a}$$

Calculate the average rate of change of the function  $f(x)$  over the domain  $\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$  if:

a  $f'(x) = 12 \sin 6x$

b  $f'(x) = 15 \cos 3x$

- 5 Fluid leaks from a storage facility and its rate of change after  $t$  days is given by:

$$F'(t) = 3000e^{-0.5t}$$

- a Find the amount of fluid that will leak in the first 7 days, correct to two decimal places.
- b Find the amount of fluid that will leak in the next 7 days, correct to two decimal places.

- 6 Water flows into, then out of, a container at a rate  $\frac{dV}{dt}$  litres per minute given by:

$$\frac{dV}{dt} = t(10 - t)$$

Where the number of minutes,  $t \geq 0$ .

- a Sketch the graph of  $\frac{dV}{dt}$ .
- b Hence, find the maximum flow rate.
- c Find an expression for the volume of water,  $V$  litres, in the container at time  $t$  minutes, assuming that the container is initially empty.
- d Find the total time taken for the container to fill and then empty.

- 7 The rate of emission  $\frac{dE}{dt}$  of CFCs in Australia, measured in tonnes per year, was given by:

$$\frac{dE}{dt} = 100 + \left( \frac{60}{1+t} \right)^2$$

Where  $t$  is the time in years after 1 September 1988.

- Find the rate of emission on 1 September 1988.
- Find the rate of emission on 1 September 1997.
- Find the value that  $\frac{dE}{dt}$  approaches as the years pass by.
- Find the total amount of CFCs emitted in Australia during the years 1988 to 1997.

- 8 Wheat is poured from a silo into a truck at a rate of  $\frac{dM}{dt}$ , measured in kilograms per second, given by:

$$\frac{dM}{dt} = 81t - t^3$$

Where  $t$  is the time in seconds after the wheat begins to flow.

- Find an expression for the mass  $M$  kg of wheat in the truck after  $t$  seconds, if initially there was 1 tonne of wheat in the truck.
- Find the total mass of wheat in the truck after 8 seconds.
- Find the largest value of  $t$  for which the expression for  $\frac{dM}{dt}$  is physically possible.

- 9 The rate of flow of water into a supply tank is given by:

$$V' = 2000 - 30t^2 + 5t^3$$

For  $0 \leq t \leq 7$ , where  $V$  is the amount of water (in litres) in the tank  $t$  hours after midnight.

- Find the initial flow rate.
- Complete the table of values.

|       |    |   |   |   |   |
|-------|----|---|---|---|---|
| $t$   | -1 | 0 | 1 | 4 | 6 |
| $V''$ |    |   |   |   |   |

- Hence, state the time  $t$  when the flow rate is a maximum, and the maximum flow rate at this time.
- Sketch the graph of  $V'$ .
- Find the area bound by the graph of  $V'$  and the  $t$ -axis between  $t = 0$  and  $t = 3$ .
- Describe what the area found in part (e) represents.

10 An ice cube with a side length of 25 cm is  removed from the freezer and starts to melt at a rate of  $25 \text{ cm}^3/\text{min}$ . Let  $V$  be its volume  $t$  minutes after it is removed from the freezer.

- a State the equation for the rate of change of volume.
- b State the equation for the volume,  $V$ , of the cube as a function of time  $t$ .
- c Find the value of  $t$  when the ice cube has melted completely.

11 The mass,  $m$  grams, of a raindrop falling for  $t$  seconds is increasing at a rate given by:

$$\frac{dm}{dt} = \frac{1}{120} \left( \sqrt{t} + \frac{t^2}{12} \right) \text{ g/s}$$

- a Given that the initial mass of the raindrop is zero, find the mass  $m$  of the raindrop after  $t$  seconds.
- b Find the exact mass of the raindrop after 16 seconds have passed.
- c Another raindrop starts as a gas particle with a mass of 0.004 g. How much heavier will it be after 16 seconds than a raindrop that is initially weightless?

12 To determine the number of staff required to serve customers at a new cafe, a consultancy company monitored the number of customers over a ten hour period.

The rate of change of the number of customers seen during  $t$  hours of service was seen to be modelled by:

$$\frac{dC}{dt} = 50\pi \cos \left( \frac{\pi(t-3)}{20} \right)$$

- a Approximately how many customers visited the cafe in the first hour?
- b Approximately how many customers visited the cafe in the last two hours?
- c Calculate the average number of customers per hour over the entire ten hour day.

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## Cost, revenue and profit applications

13 The marginal cost for the production of the  $x$ th item is modelled by:

$$C'(x) = 8x + 909 \text{ dollars per item}$$

- a Find the net change in cost for producing between 13 and 19 items.
- b Find the average change in cost for producing between 13 and 19 items.

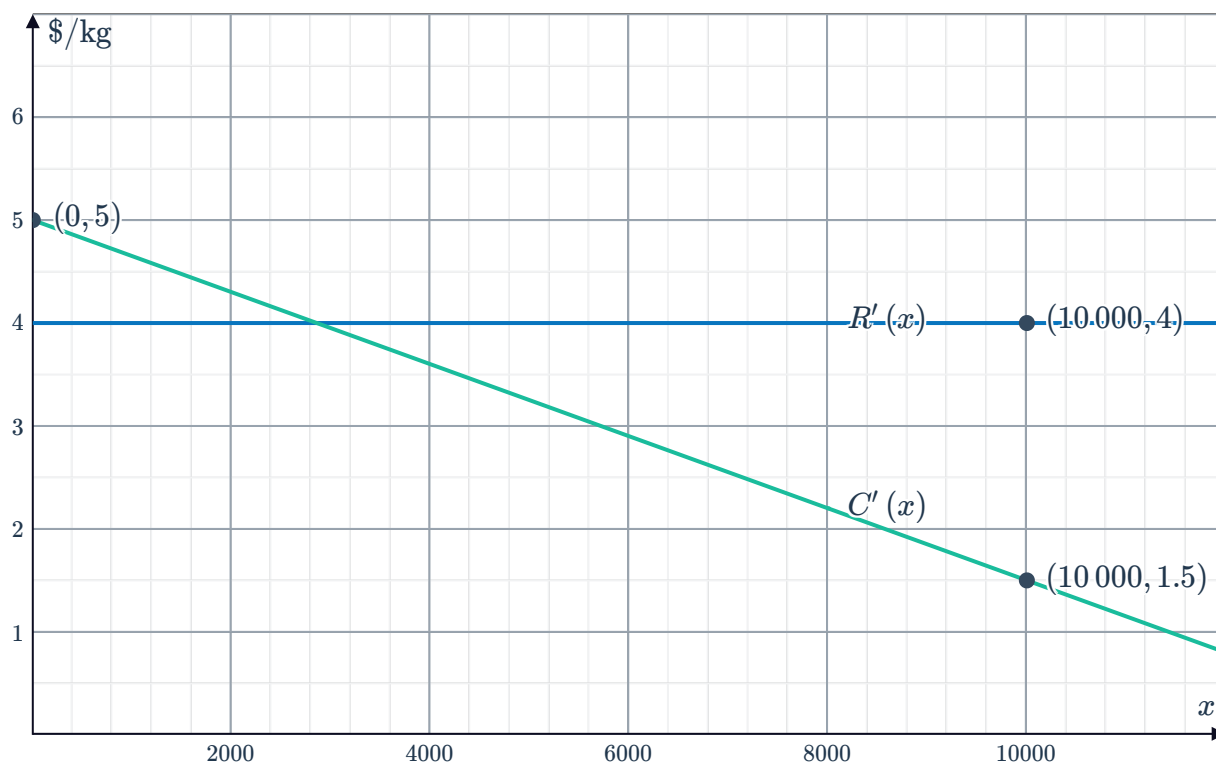
- 14 The marginal profit from the sale of the  $x$ th item is given by:

$$P'(x) = 0.0012x^2 + 1.2x - 4.1$$

Where  $P(x)$  is the profit from selling  $x$  items.

- Given that the company incurs a loss of \$60 if no items are sold, find an expression for  $P$  in terms of  $x$ .
- Hence, determine the profit from selling 70 items.
- Find the net change in profit if the number of items sold changes from 70 to 140 items.

- 15 The marginal cost,  $C'(x)$ , and marginal revenue,  $R'(x)$ , of a company producing  $x$  kg of coffee is plotted below:



- Find the area under each of the following functions over the interval  $[0, 10\,000]$ :
  - $R'(x)$
  - $C'(x)$
- Hence, state the total revenue of producing the first 10 000 kg of coffee.
- Hence, state the total cost of producing the first 10 000 kg of coffee, given initial set-up costs of \$2000.
- The profit for selling  $x$  kg of coffee is given by  $P(x) = R(x) - C(x)$ . Determine the total profit of the first 10 000 kg of coffee.
- Market changes result in the marginal cost of coffee increasing by 20% while the marginal revenue decreases by 10%. Determine the total profit of the first 10 000 kg of coffee after the market change.

- 16**  $C(x)$  is the cost of producing  $x$  items of a  product. The marginal cost of producing the  $x$ th item is given by:

$$C'(x) = 0.4x^3 + 4x$$

- a** Determine the cost function  $C(x)$ , in terms of  $x$ , given the fixed cost of production is \$2000.
- b** Hence, determine the cost of producing 50 items.

- 17**  $C(x)$  is the cost of producing  $x$  units of a certain product. The marginal cost of producing the  $x$ th unit is given by:

$$C'(x) = 9x^2 - 70x + 500$$

- a** Determine the cost function  $C(x)$  in terms of  $x$ .
- b** Hence, determine the extra cost incurred by producing 60 units rather than 20 units.

- 18**  $C(x)$  is the cost of producing  $x$  units of a certain product. The marginal cost of producing the  $x$ th unit is given by:

$$C'(x) = \frac{120}{\sqrt{x}}$$

- a** Determine the cost function  $C(x)$  in terms of  $x$ .
- b** Hence, determine the extra cost incurred by producing 81 units rather than 25 units.

- 19**  $R(x)$  is the revenue from producing and selling  $x$  items. The marginal revenue from producing and selling the  $x$ th item is modelled by:

$$R'(x) = 2.7x(x - 200)$$

- a** Determine the change in revenue from producing and selling the 41st item.
- b** Determine the net change in revenue from selling 50 items to selling 80 items.
- c** Determine the average change in revenue from selling 50 items to selling 80 items.

- 20** A factory manufactures tractors. The marginal cost,  $C'(x)$ , and the marginal revenue,  $R'(x)$ , are given by:

$$C'(x) = 3x + 1, R'(x) = 9x + 6x^2 + 6$$

- a** Given that zero production results in zero profit, determine the profit function  $P(x)$ .
- b** Hence, determine the total profit from production and sale of the first 30 tractors.

- 21**  $C(x)$  is the cost of producing  $x$  units of a certain product. The marginal cost of producing the  $x$ th unit is given by:

$$C'(x) = \frac{440}{(x+5)^2}$$

- a** Determine the extra cost incurred by producing 30 items rather than 20 items.
- b** Determine the extra cost incurred by producing 50 items rather than 30 items.

- 22**  $R$  is the total revenue, in thousands of dollars, from producing and selling a new product,  $t$  weeks after its launch. The marginal revenue from producing and selling the  $x$ th item is modelled by:

$$\frac{dR}{dt} = 401 + \frac{500}{(t+1)^3}$$

- a** Given that the initial revenue at the time of launch was zero, state the revenue function.
- b** Find the average revenue earned over the first 5 weeks.
- c** Calculate the revenue earned in the 6th week.

- 23** For each of the following marginal profit functions, which give the profit, in dollars, of producing and selling  $x$  items of a product. Find:

- i** The extra profit associated with producing and selling the 101st item.
- ii** The net change in profit from producing and selling the first 100 items.
- iii** The average profit from producing and selling the first 100 items.

**a**  $P'(x) = 30\pi \cos\left(\frac{\pi x}{600}\right)$

**b**  $P'(x) = 20\pi \sin\left(\frac{\pi x}{600}\right)$